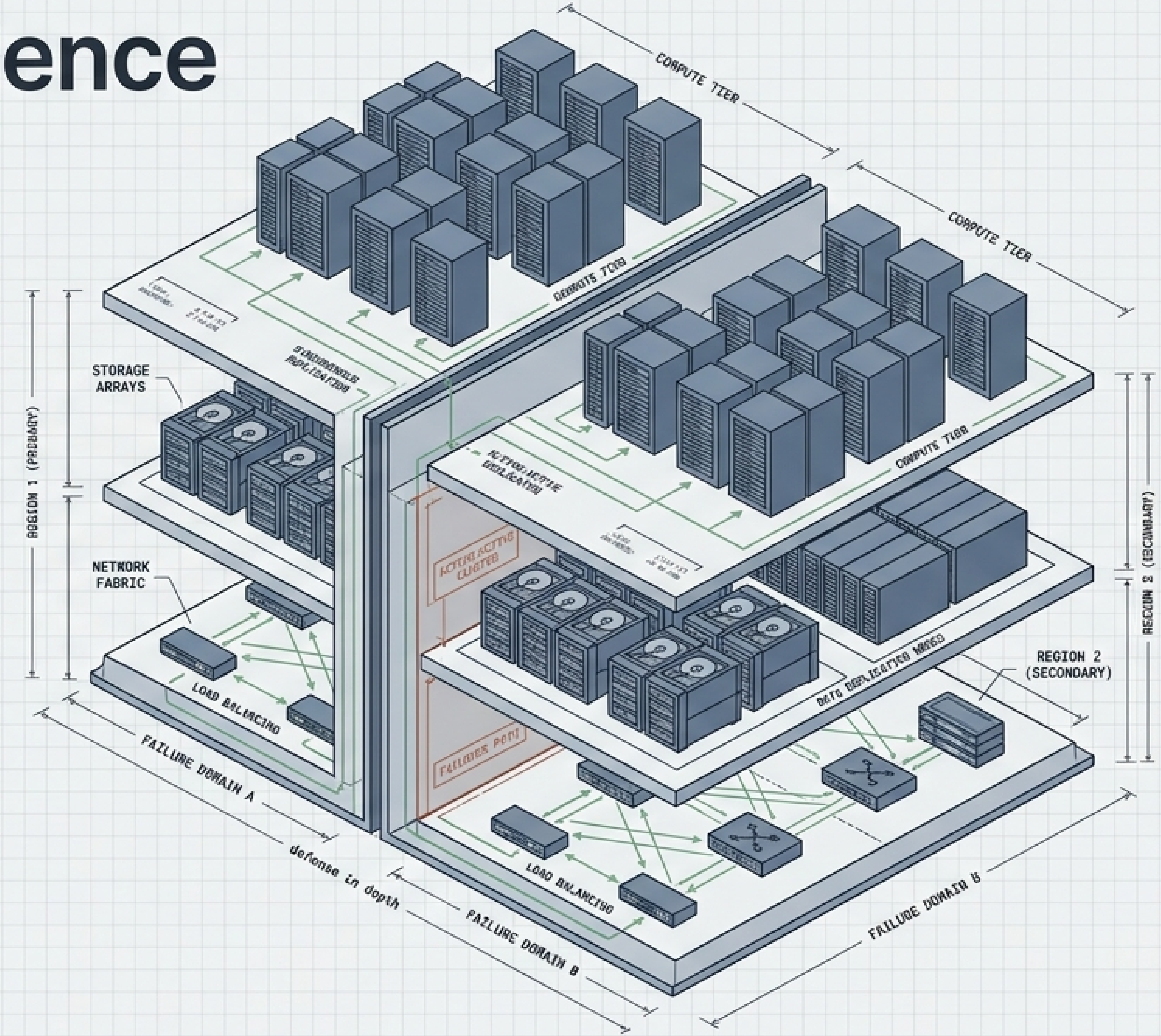
Architecting Resilience

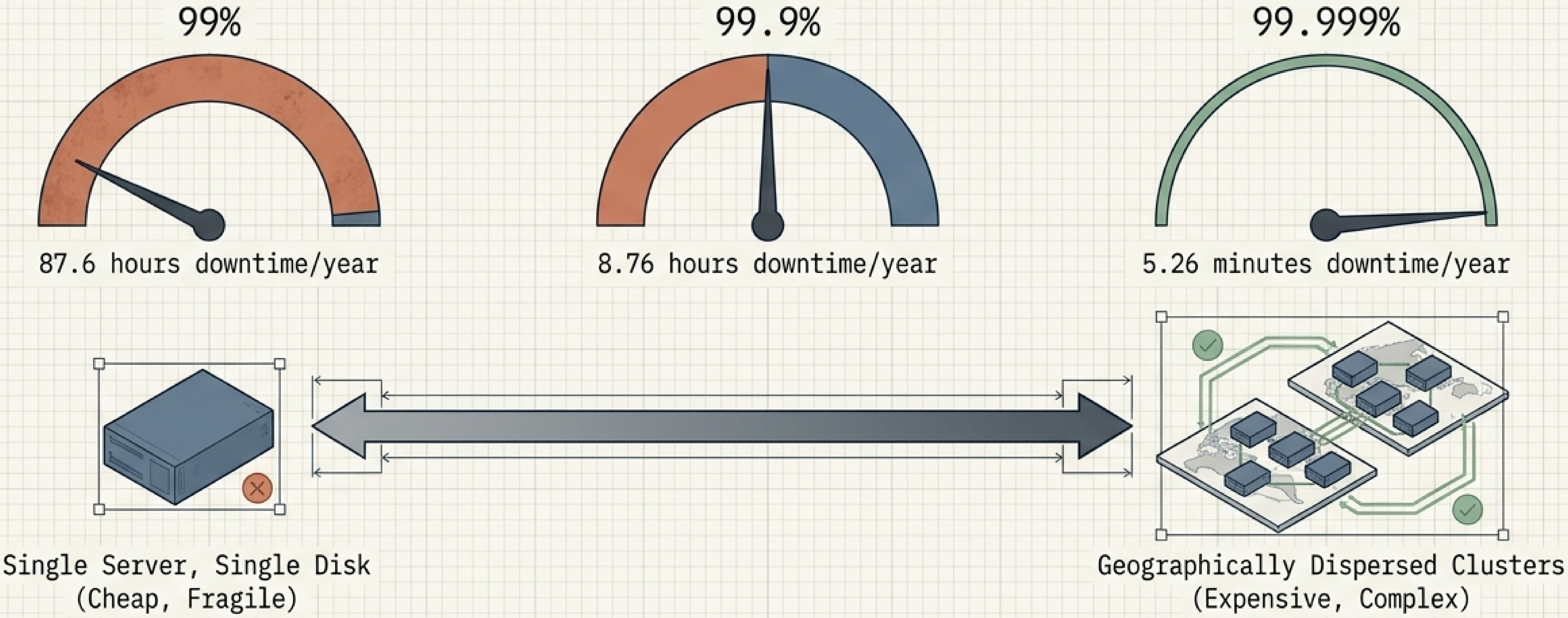
High Availability and Disaster Recovery



SYSTEM STATUS: NORMAL | AVAILABILITY: 99.999%

LAST SACRUP: 2834-07-25 88:08 UTC
RECOVERY TIRE 26JECT4VE (9TO): < 5 RIN
RECOVERY POIAT OBJECTIVE (9PB): 0 RIA

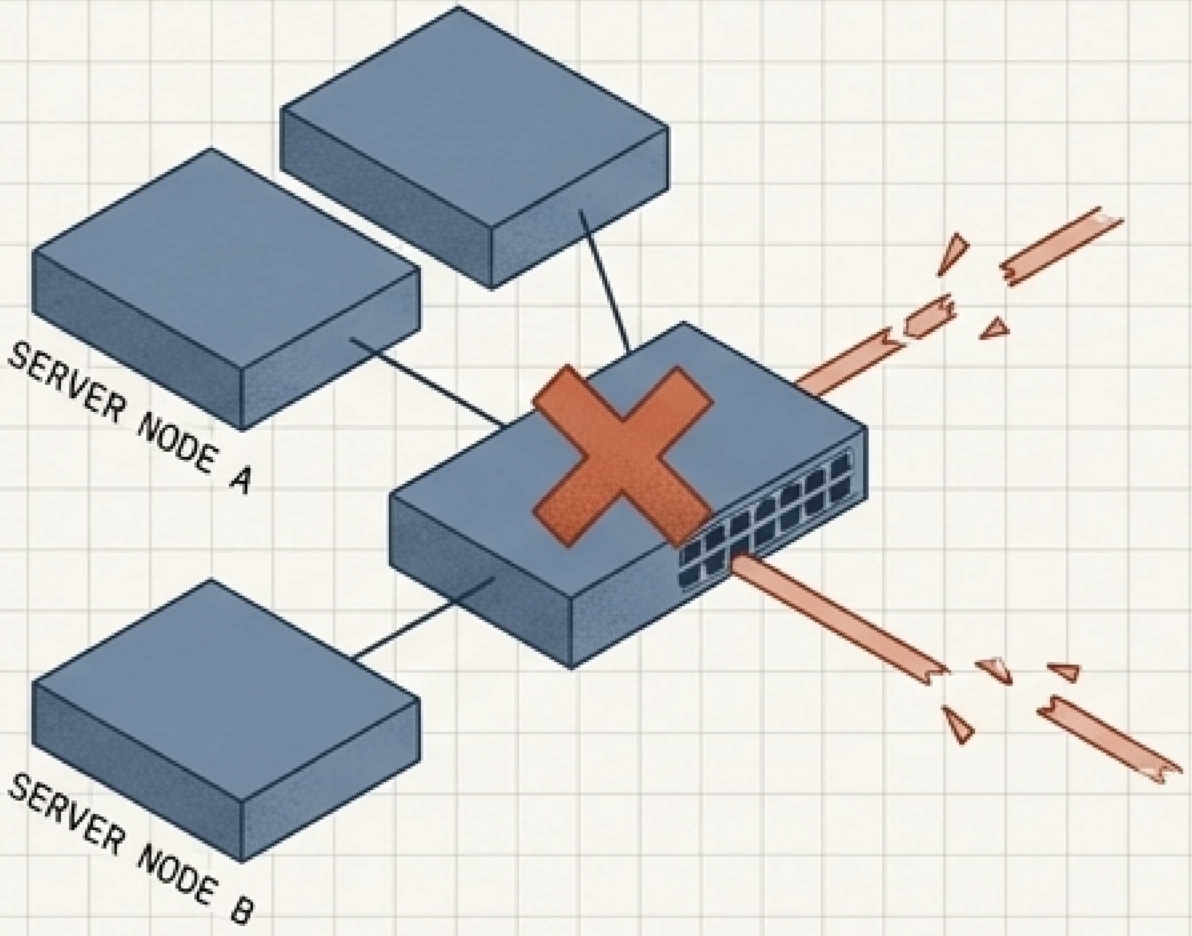
The Availability Spectrum



> The engineering question is not “How can I make the system impossible to break?” The question is: “Which failures must the system survive automatically, and which require deliberate recovery?”

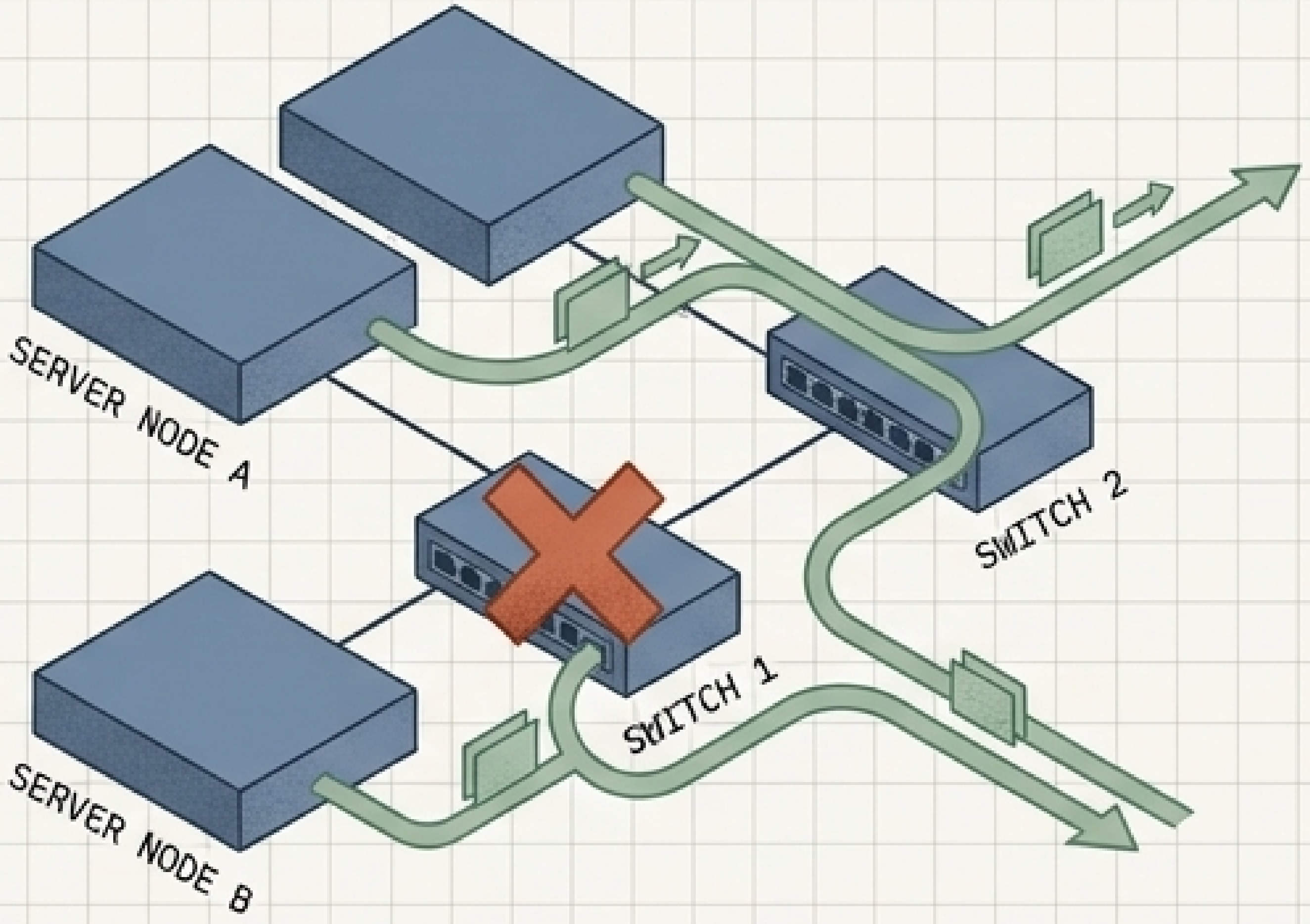
Eliminating Single Points of Failure

Single Point of Failure (SPOF)



Redundant servers, single network. The switch is a SPOF. Traffic stops entirely.

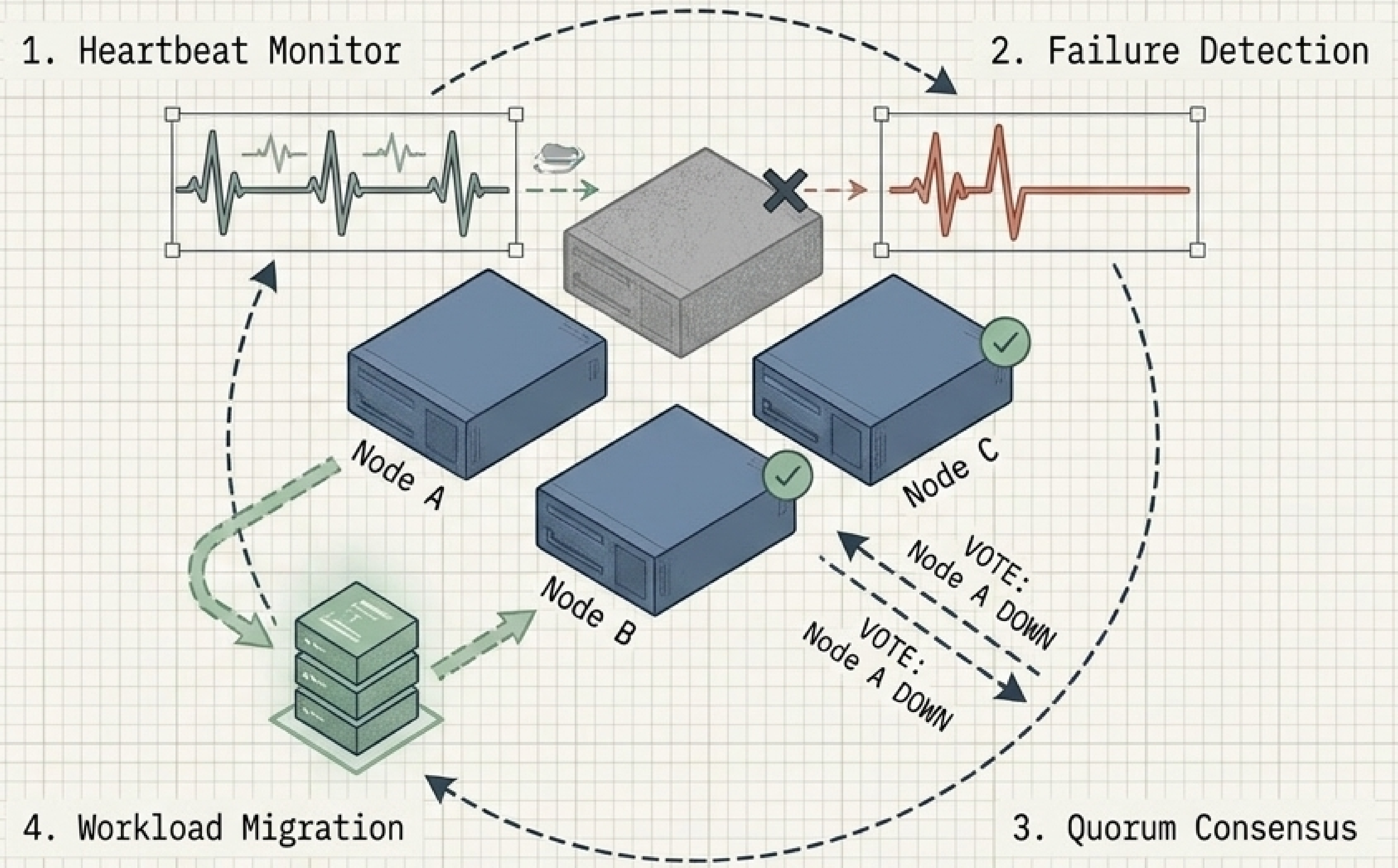
Redundancy without SPOF



Redundant servers, dual network. Traffic routes around the failure.

CONSTRAINT: Redundancy is useless if components share the same Failure Domain.
Two power supplies connected to one failing power grid provide no resilience.

Layer 1: High Availability and Automatic Failover



MTBF (Mean Time Between Failures)

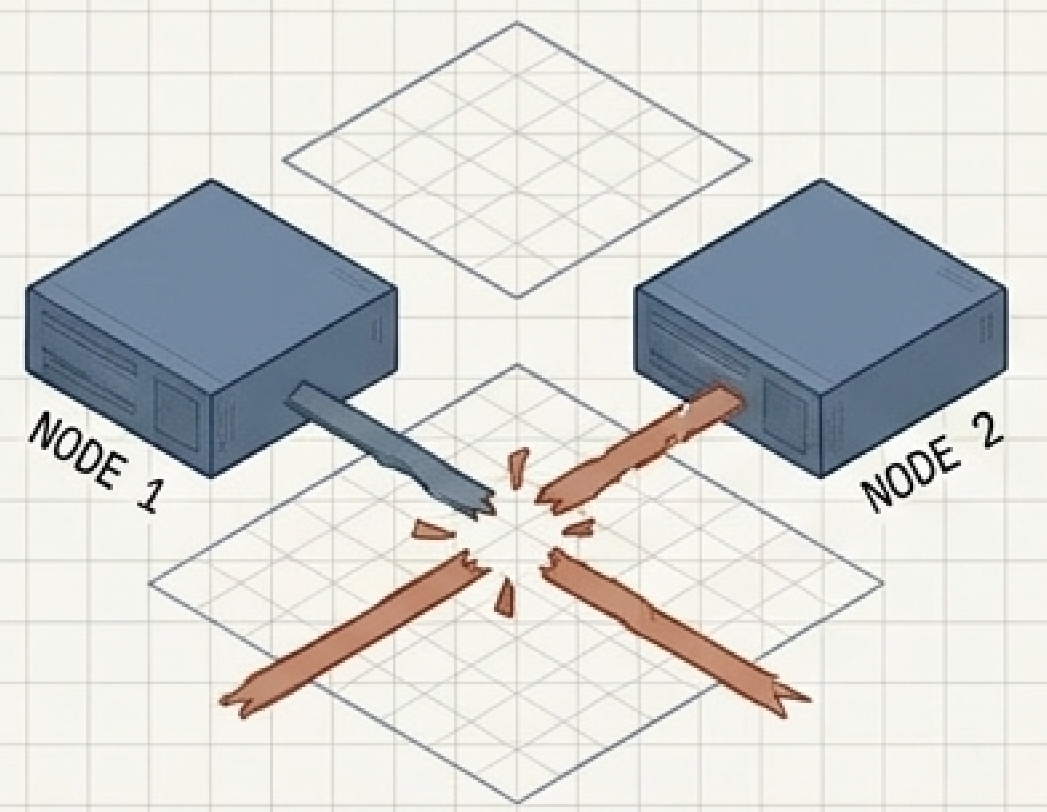
Extend this metric to improve overall system reliability.

MTTR (Mean Time To Repair)

Shrink this metric through automatic failover to improve availability.

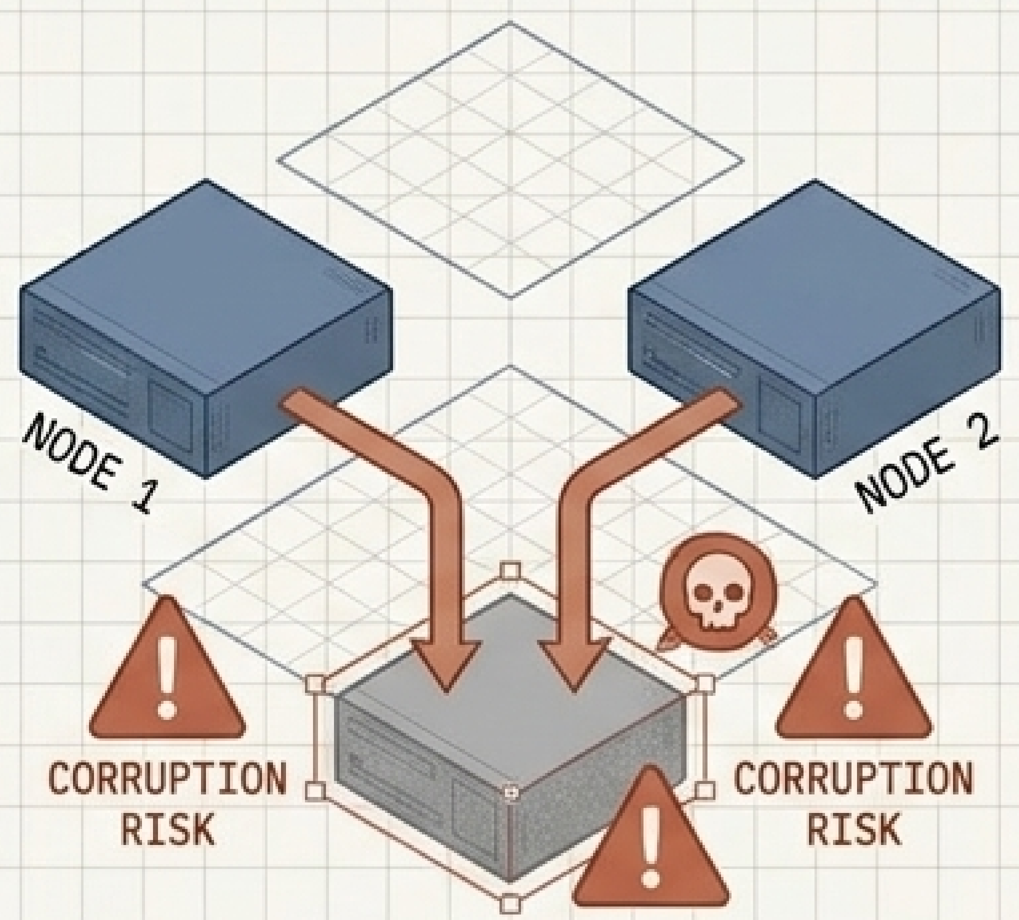
The Danger of Split-Brain

1. The Break



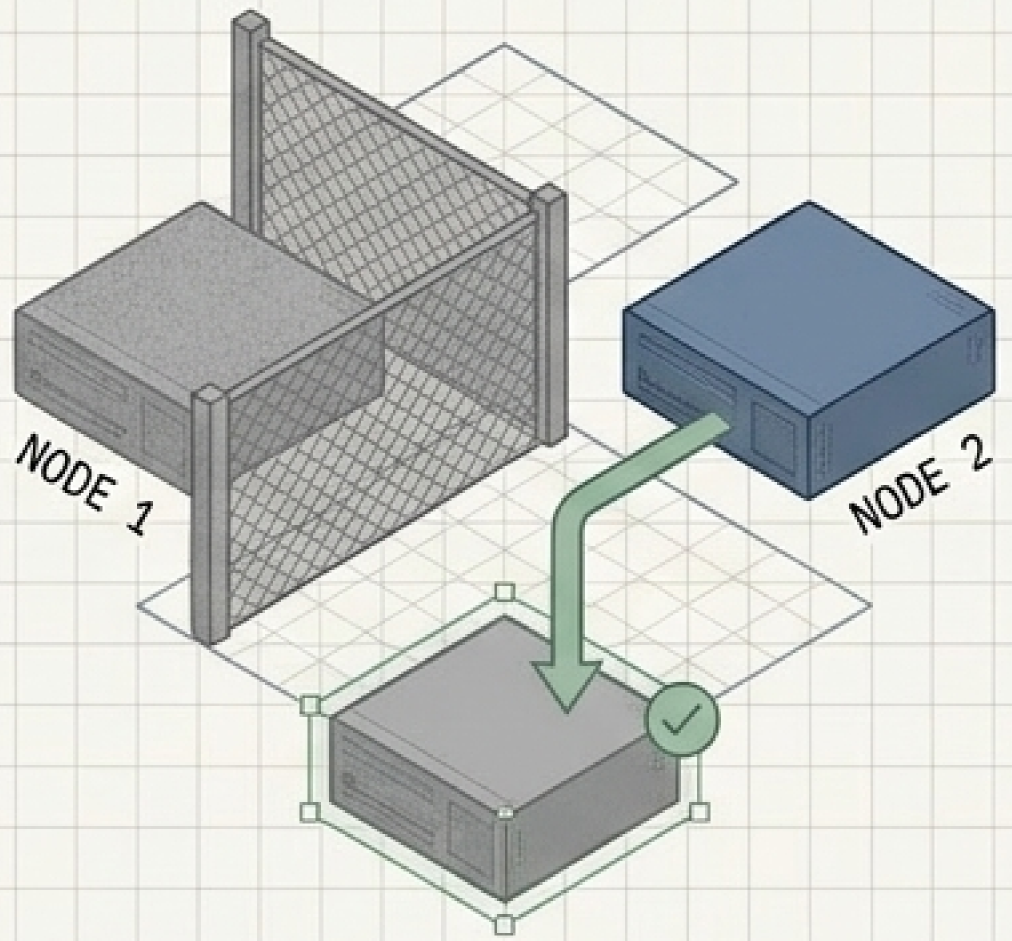
Network connection is lost. Both nodes remain operational but cannot communicate.

2. The Threat



Both nodes assume the other is dead. Both attempt to write data. Result: Severe Corruption.

3. The Fence

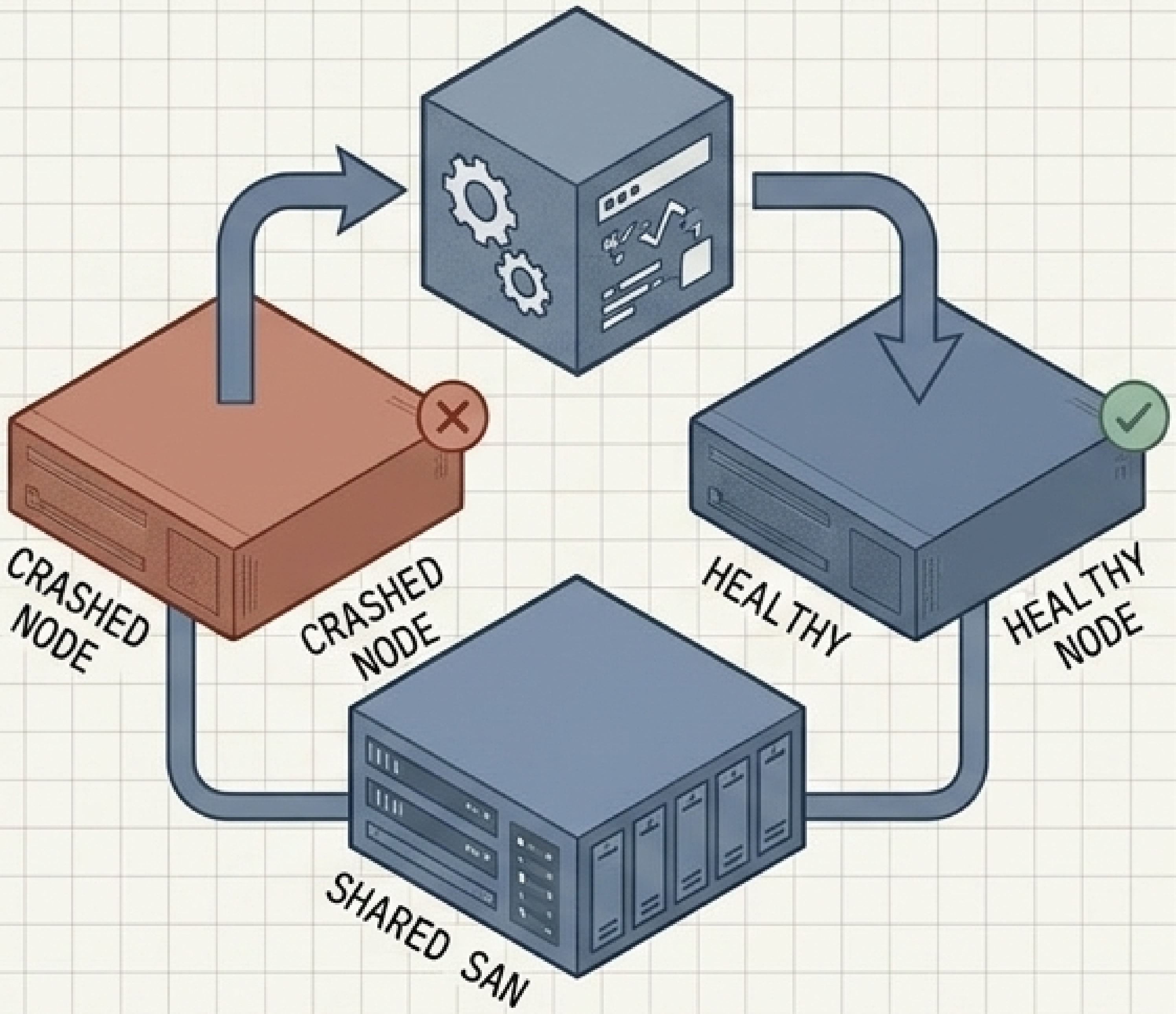


A physical or logical “fence” isolates Node 1, preventing it from accessing shared resources.

RULE: Automatic failover without reliable fencing is more dangerous than no automatic failover at all.

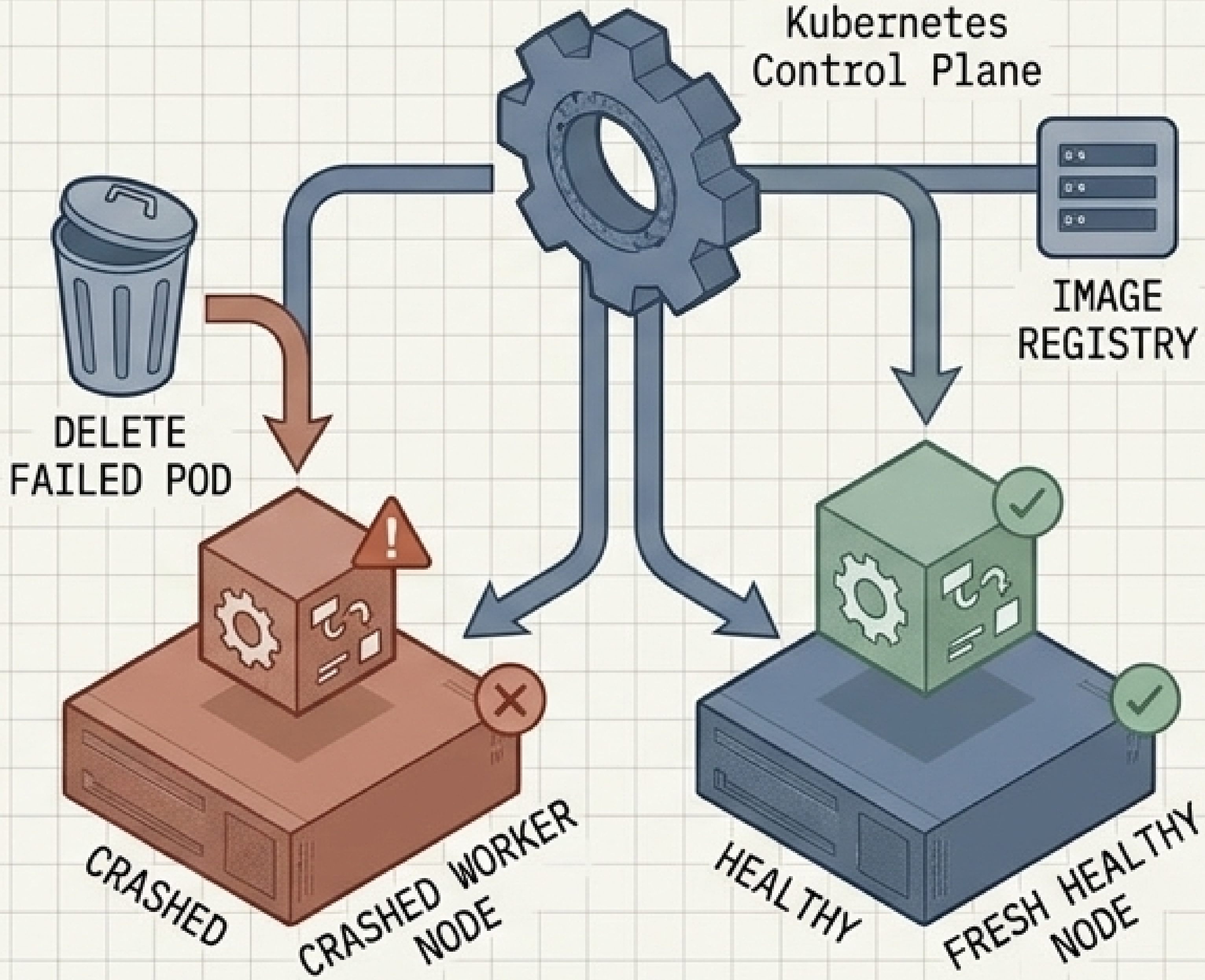
Workload Recovery: Virtual Machines vs. Containers

Stateful Preservation (VM HA)



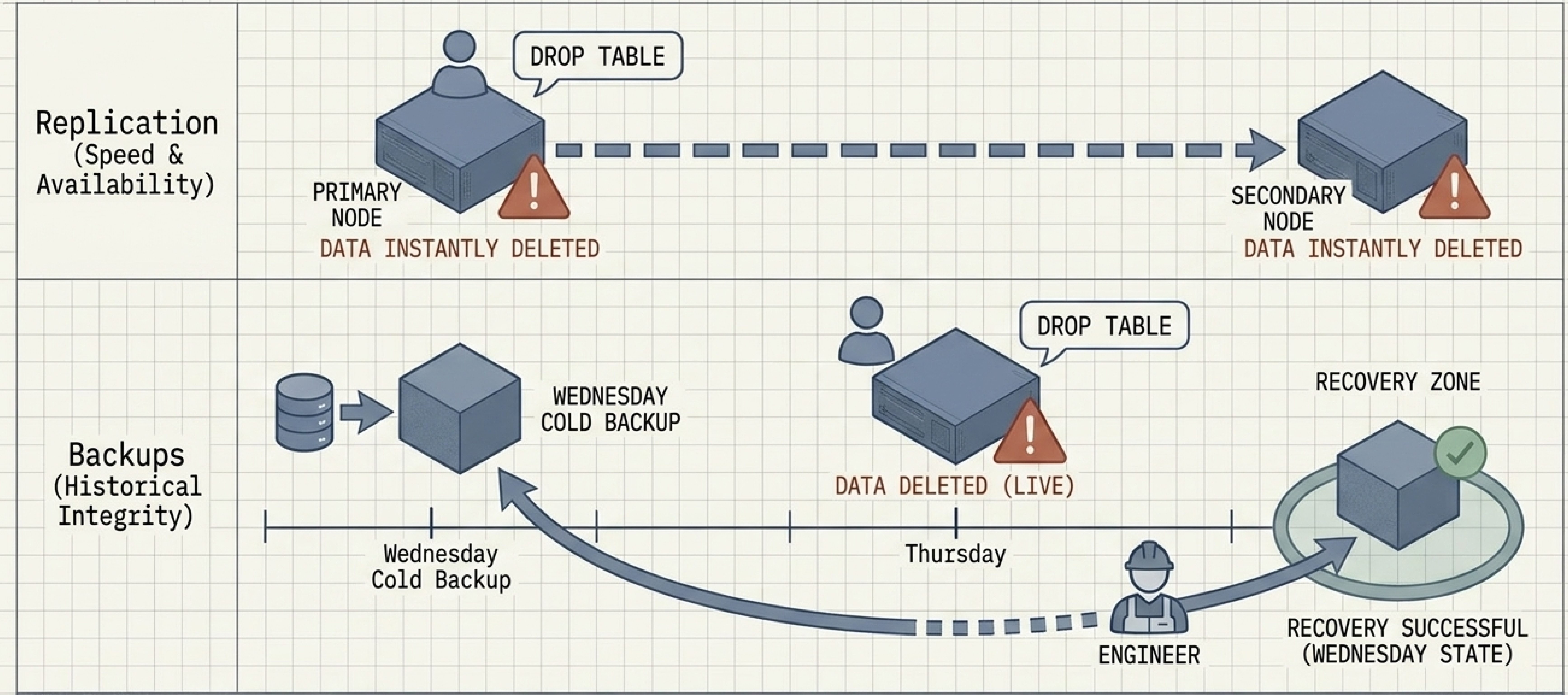
Preserves the exact state of the machine as the unit of recovery. Requires shared storage to reconnect the virtual disk.

Disposable Replacement (Kubernetes)



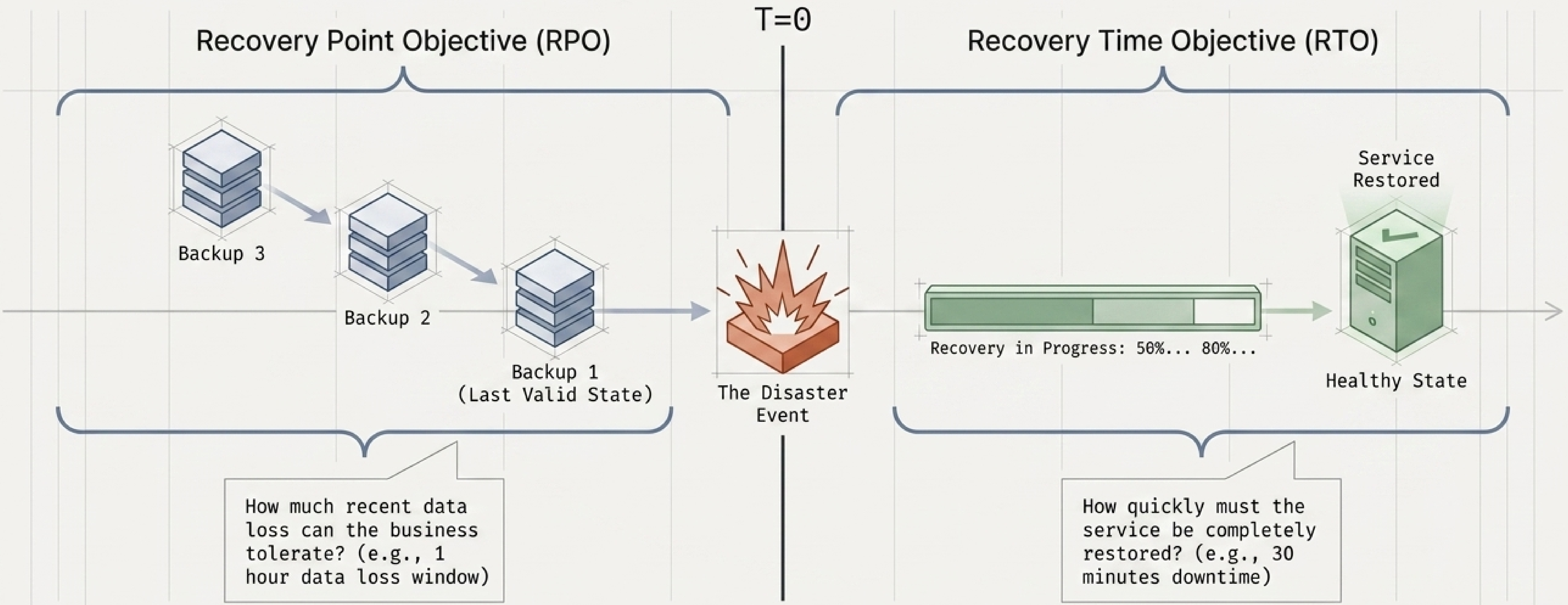
Treats Pods as ephemeral and disposable. Deletes the failed instance and spawns an identical clone, recovering state via Persistent Volumes.

Replication is for Speed. Backups are for History.



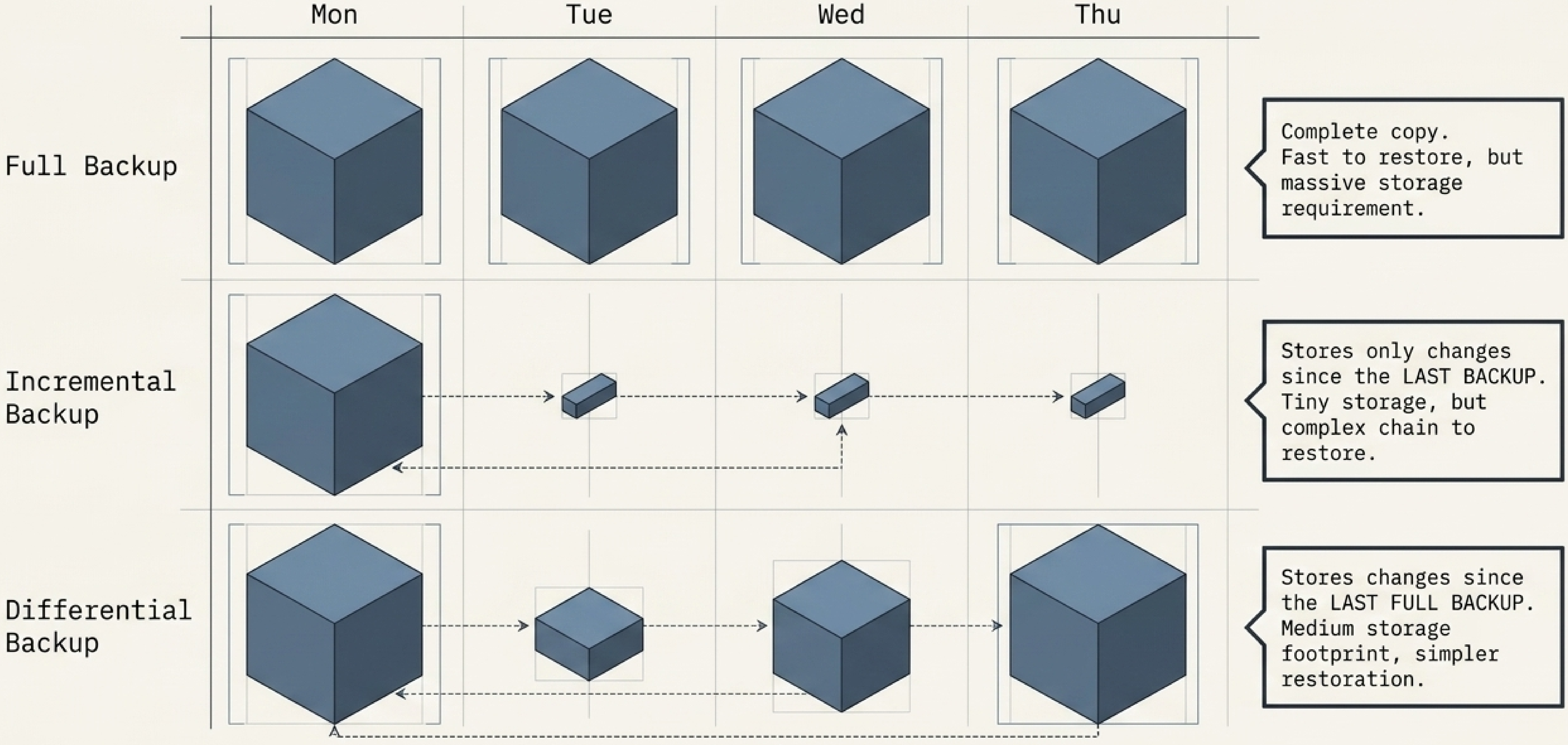
Replication faithfully reproduces your mistakes instantly.
Backups provide a historical recovery point to undo them.

The Metrics of Disaster: RPO and RTO

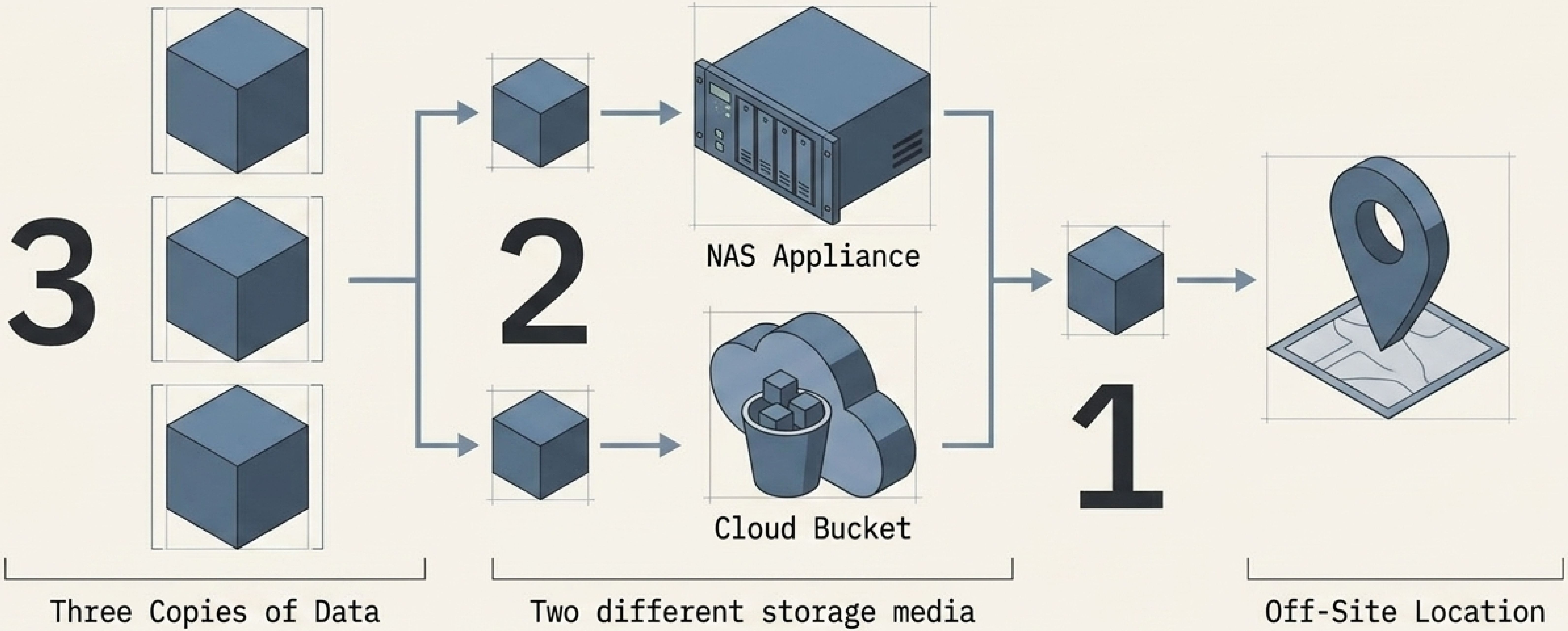


These values must come from business requirements, not arbitrary technical preferences.

Structuring the Backup Chain

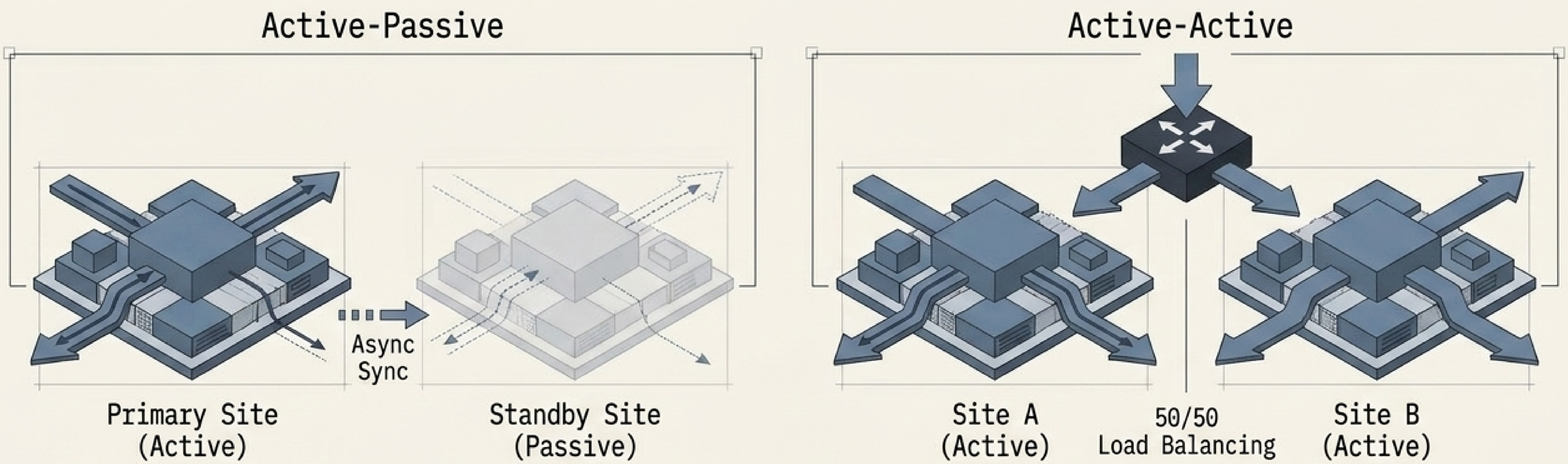


The 3-2-1 Backup Imperative



SECURITY IMPERATIVE: In the ransomware era, the '1' off-site copy must be offline and immutable. A backup an attacker can easily delete is not a reliable recovery copy.

Layer 3: Disaster Recovery Sites



Cold Site	Warm Site	Hot Site
Infrastructure space only. No hardware or data pre-staged. • Lowest Cost, Slowest RTO	Hardware available and data periodically synced. • Medium Cost, Moderate RTO	Fully replicated, active standby ready for instant activation. • Highest Cost, Fastest RTO

Gemini Notebook

The Recovery Dependency Chain

Web & App Servers

Stateless frontend. Must connect to a healthy, restored database to function.

↑

Database Clusters

Stateful data tier. Must achieve quorum before accepting connections.

↑

Active Directory & DNS

Identity and routing.
(Account for DNS TTL propagation delays).

↑

Network & VPN Infrastructure

The physical and logical base layer.

MTTR depends on sequence.

A successful VM boot is not successful application recovery. Restoring a web app before its database guarantees failure.

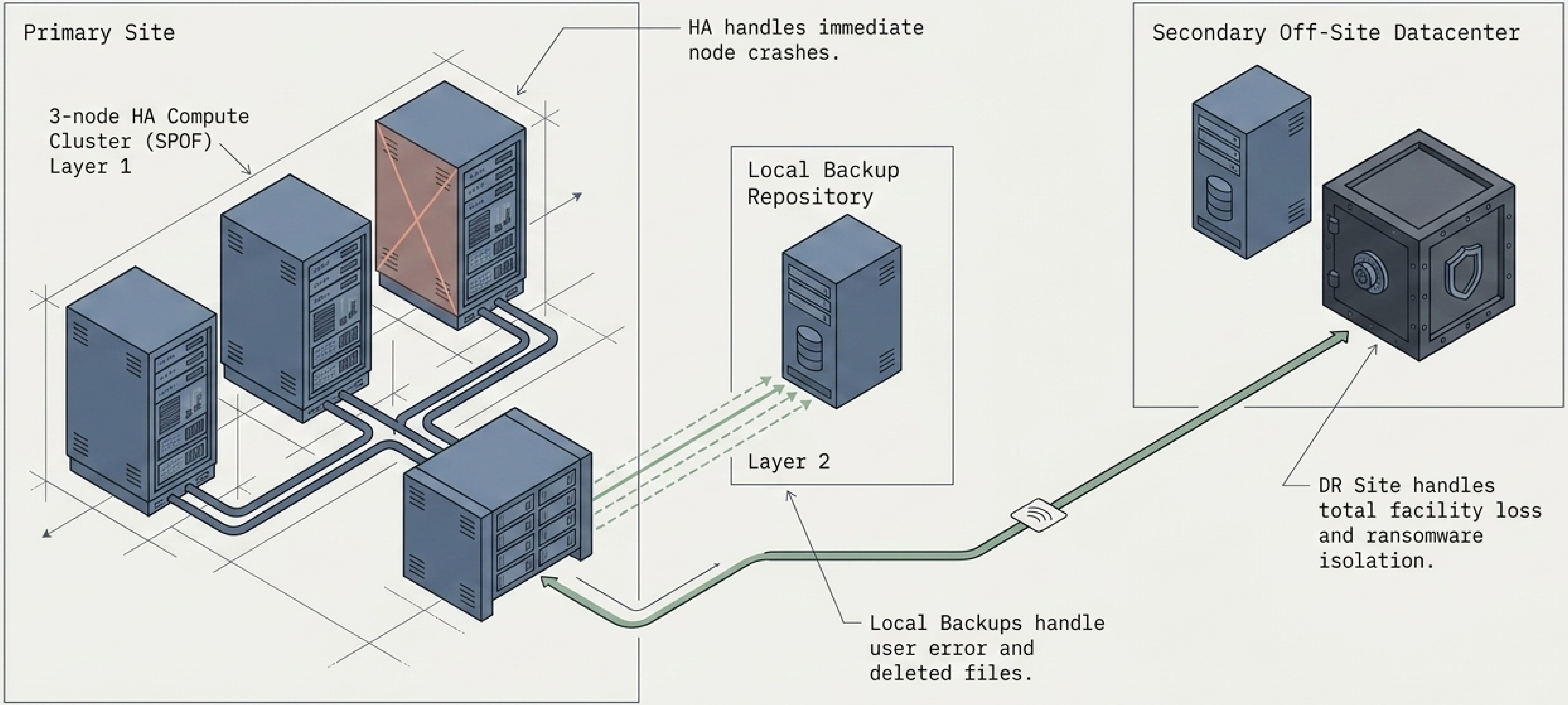
Document the exact recovery sequence in the DR Runbook.

The Resilience Diagnostic Matrix

Failure Scenario	High Availability (HA)	Backup	Disaster Recovery (DR)
Host / Node Failure	Solves	Partial	Fails
Accidental File Deletion	Fails	Solves	Fails
Ransomware / Corruption	Fails (Keeps corruption online)	Solves (If immutable)	Partial
Datacenter Destruction	Fails	Partial (If off-site)	Solves

Conclusion: No single technology provides **complete resilience.** Different technologies protect against completely different failure classes.

Unified Defense-in-Depth Architecture



The Engineer's Mandate

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> Do not ask only whether a system can survive a failure.
> Ask WHICH failures it can survive.
> Ask HOW QUICKLY it recovers (RTO).
> Ask HOW MUCH DATA can be lost (RPO).
> Ask what happens when redundancy itself fails.
> And ask whether the recovery procedure has ACTUALLY
  BEEN TESTED.
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A backup is just an assumption until it has been restored successfully.
Measure whether recoverable data exists, not just if the job ran.